BFS vs DFS – Time Complexity, Space Complexity & Test Cases

Based on the uploaded SLE-2 BFS vs DFS Profiling Graph report.

1. Time Complexity

Algorithm	Time Complexity
BFS	O(V + E)
DFS	O(V + E)

V = number of vertices/nodes and E = number of edges/connections. For a fixed-neighbour grid/maze, this is commonly treated as approximately O(V).

2. Space Complexity

Algorithm	Space Complexity
BFS	O(V)
DFS	O(V)

BFS uses a queue and may store many frontier nodes. DFS uses a stack/recursion and stores nodes along the current search path.

3. Test Cases

Test Case	Input / Condition	Expected Result
TC1	Small maze with direct path	BFS and DFS find the goal.
TC2	Maze with multiple paths	BFS finds the shortest path; DFS may find a longer path.
TC3	Large maze	Compare execution time and nodes expanded.
TC4	Goal close to start	Both terminate quickly.
TC5	Goal far from start	More nodes may be explored.
TC6	No path to goal	Both report that no path exists.
TC7	Start = Goal	Both return path length 0.
TC8	Maze with dead ends	Compare unnecessary exploration by BFS and DFS.

4. Results from the Uploaded Report

Metric	BFS	DFS
Average Time	0.3268 ms	0.1998 ms

Nodes Expanded	328	225
Path Length	39 steps	45 steps

Conclusion: For the maze in the uploaded report, DFS took less average time and expanded fewer nodes, while BFS found the shorter path.